Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · GenAI · LangChain · Financial Data Pipelines · REST APIs · AWS · Regulated Industries · 9yr Production

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PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience delivering scalable, secure applications through the full development lifecycle - design, coding, automated testing, CI/CD, deployment, and production support - with deep specialization in financial data pipelines, generative AI integration, and regulated industry compliance. At Entrata, I owned the HUD federal compliance pipeline for 7 years: Python and AWS-powered data processing for 180,000 domain records that drive eligibility and payment calculations for 100K+ housing units, maintained with 100% PHI/HIPAA compliance, zero security incidents, and full auditability across 7yr of production operation - the financial services reliability and compliance standards Citi requires. I build and ship generative AI applications in production: AgentProbe is a multi-provider LLM platform (OpenAI API, Anthropic API) with MLOps evaluation framework, prompt management, CI-integrated regression gate, and RAG infrastructure; LeaseI is a production AI system with human-in-the-loop controls, audit logging, and compliance-grade guardrails. My primary stack is Python + FastAPI + AWS + Docker with LangChain/LangGraph for generative AI workflows. I apply software engineering best practices throughout - version control, automated testing, code reviews (300+), and CI/CD with GitHub Actions. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr), TypeScript, JavaScript, PHP, SQL, PL/SQL, FastAPI, Flask, REST APIs, LangChain, LangGraph, React.js, HTML/CSS, jQuery, Bash

Generative AI & MLOps: Generative AI, LLMs, OpenAI API, Anthropic API, Ollama, Hugging Face, Prompt Engineering, Prompt Management, MLOps, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, NLP, BERT, spaCy, PyTorch, scikit-learn

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (async/event-driven), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Automated Testing, Security & Encryption, PHI/HIPAA Compliance, ETL Pipelines, Performance Engineering, Agile/Scrum, Linux, Git

PROJECTS

AgentProbe: Generative AI Platform - MLOps, Evaluation & REST API

GenAI · MLOps · LLMOps · Python · FastAPI

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · OpenAI API · Anthropic API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production generative AI platform integrating multiple LLM providers (OpenAI API, Anthropic API, Ollama) via clean REST API abstractions - built with MLOps best practices from day one. **REST API layer:** FastAPI backend with 15 endpoints; multi-provider LLM routing without prompt rewrites; clean separation of concerns. **MLOps evaluation framework:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency) with prompt management, model monitoring, and CI-integrated regression gate (GitHub Actions) that blocks quality degradation before deployment. **RAG infrastructure:** ChromaDB vector database, chunking optimization, embedding selection. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable, fully documented.

LeaseI: Production AI System - Compliance-Grade & Human-in-the-Loop

GenAI · HITL · Regulated · Security

Python · LangGraph · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable, secure generative AI application built through the full software development lifecycle - automated document processing (OCR, section-aware extraction), LangChain-powered AI analysis with human-in-the-loop escalation for high-stakes regulated decisions, and end-to-end test coverage with CI regression gate. Compliance-grade design: deterministic guardrails ensure AI never auto-resolves regulated decisions without human approval; full audit logging for every decision path; encrypted data handling throughout. 93.6% precision across 1,240 test cases; zero unsupported-claim rate on production eval set.

AI on FHIR: Cloud-Native Application - HIPAA-Compliant Enterprise APIs

Cloud-Native · HIPAA · REST APIs · Security

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant cloud-native application integrating AI/ML with enterprise REST APIs - LLM-powered NLP translates natural-language queries into structured FHIR API calls; RBAC with authentication and full data isolation; ~60% reduction in manual retrieval time; containerized with Docker. Demonstrates the regulated financial services integration pattern: secure application development, compliance-by-design, and auditability built in from the start.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production GenAI Application Delivery

- Designed and delivered a production generative AI application (Python, LangGraph, FastAPI) for an external customer end-to-end - gathered requirements from stakeholders, architected the solution with RAG over regulatory documents, human-review guardrails, and REST API layer; deployed to production in weeks as sole technical owner with full documentation and automated testing.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering best practices; translated complex technical outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Financial data pipelines & ETL at production scale** - owned the HUD federal compliance pipeline for 7 years (Python, AWS Glue, S3; federal income-limit data → 180,000 domain records powering eligibility and payment calculations for 100K+ housing units); annual releases with baseline validation, reconciliation, and rollback; zero compliance violations across 7yr. Built automation pipelines eliminating 80% of manual operational workload; 200+ SQL optimizations (30% performance gain).
- **Secure, compliant, regulated application development** - 100% PHI/HIPAA compliance with Libsodium-encrypted data pipelines; zero security incidents across 7yr of production; AWS S3/EC2/RDS/Glue; Docker microservices (50% scalability improvement); RabbitMQ event-driven async with retry logic and idempotency; Redis caching; 99.5% uptime across multi-tenant SaaS.
- **Software engineering best practices & mentoring** - GitHub Actions CI/CD; automated testing throughout; 300+ peer code reviews; internal debugging tools reducing triage time 40%; 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules shipped; mentored 5+ engineers; leveraged generative AI tools (Claude Code) to accelerate delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; ETL data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017